FMCG Multi-Country Sales Dataset Analysis

(MySQL Analysis)

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4.1 Analytical Questions

Now that we've understood the dataset, performed data quality check and prepared the dataset for further analysis, its time to think about the questions that we're going to work on in this project –

Sales Performance & Growth

1. What are the overall revenue and sales-volume levels of the business?
2. How does revenue and sales volume change over time?
3. What is the month-over-month sales growth?
4. What is the year-over-year sales growth?

Product & Category Performance

5. Which are the top 20 products/SKUs by revenue?
6. Which products generate the highest sales volume?
7. Which categories and brands contribute the most revenue?
8. Which products and categories are underperforming?

Geographic & Store Performance

9. Which countries generate the highest revenue?
10. Which cities and stores are the strongest and weakest performers?
11. Which markets have high revenue but weak growth?

Sales Channel Performance

12. Which sales channels generate the highest revenue and sales volume?
13. How does category/brand performance differ across sales channels?

Promotion & Pricing

14. How do promotional and non-promotional sales compare?
15. Does a higher discount percentage appear to be associated with higher sales volume?
16. Which products or channels are most dependent on promotional activity?

Profitability

17. Which products generate the highest margins?
18. Which products generate high revenue but relatively weak margins (<38%)?

Inventory & Supply Chain

19. Which products experience the highest stockout exposure?
20. Is stockout exposure concentrated among particular stores & cities?
21. Which products combine high demand with high volatility?
22. How do demand classifications (ABC) and demand-volatility classifications (XYZ) compare?

Management & Strategic Insights

23. Where are the largest commercial growth and improvement opportunities?

4.2 Logical Answers to Analytical Questions

Q1. What are the overall revenue and sales-volume levels of the business?

Ans: Show the sum of (units_sold, gross_sales, net_sales) from 'sales_analysis' with no limit. This will show to sum of 1.1M rows of each field, indicating total sales volume of entire timeline. And to calculate gross revenue, find the sum of [net_sales – (units_sold * purchase_cost)]

Q2. How does revenue and sales volume change over time?
Ans: Select the YEAR and MONTH portion from 'sales_date' field, find the sum of (units_sold, net_sales) and then GROUP BY year and month, it will return total (12*3) or 36 rows of cumulative record.

Q3. What is the month-over-month sales growth?
Ans: The result obtained from Q2 has the list of 36 months sales record. Now to find growth rate, do : {(latest month record – previous month record) / previous month record} * 100%

Q4. What is the year-over-year sales growth?
Ans: Same as Q3, here only select YEAR portion from 'sales_date'.

Q5. Which are the top 20 products/SKUs by revenue?
Ans: Select Product/SKU related fields, find sum of (net_sales, units_sold) and GROUP BY product/sku fields, ORDER BY sum of net_sales and set LIMIT 20 DESC,

Q6. Which products generate the highest sales volume?
Ans: Same as Q6, just ORDER BY sum of units_sold.

Q7. Which categories and brands contribute the most revenue?
Ans: Select category, sum of (net_sales, units_sold), GROUP BY category and ORDER BY sum of net_sales DESC. Do the same to find for brands.

Q8. Which products and categories are underperforming?
Ans: Same as Q7, just ORDER BY sum of net_sales ASC. This will show the bottom performers.

Q9. Which countries generate the highest revenue?
Ans: Same process like Q7, select country and so on.

Q10. Which cities and stores are the strongest and weakest performers?
Ans: Same process as Q7, just SELECT city, store_id and GROUP BY city, store_id then ORDER BY sum of net_sales DESC (for strong performers)/ ASC (for weak performers).

Q11. Which markets have high revenue but weak growth?
Ans: Here, market means country. Just like Q7 select country, YEAR from 'sales_date' and then GROUP BY country, year also ORDER BY country, year. This will give yearly records of each country. Then calculate growth rate like in Q3.

Q12. Which sales channels generate the highest revenue and sales volume?
Ans: SELECT channel, sum of (units_sold, net_sales), average of (margin_pct), GROUP BY channel and ORDER BY sum of net_sales DESC.

Q13. How does category/brand performance differ across sales channels?
Ans: Same process like Q12. Select category once with channel and then in another query, select brand with channel. Also group them accordingly.

Q14. How do promotional and non-promotional sales compare?
Ans: Select promo_flag, sum of (net_sales, units_sold), average of (discount_pct, margin_pct) and GROUP BY promo_flag. This will return only two rows (promo_flag = 0 means non-promo, promo_flag = 1 means promotion) but it holds the sum of records of entire 1.1M dataset.

Q15. Does a higher discount percentage appear to be associated with higher sales volume?
Ans: SELECT the discount bands in CASE functions, relevant sums to see, then GROUP BY those discount bands. This will show sales volume with respect to different types of discount levels.

Q16. Which products or channels are most dependent on promotional activity?
Ans: SELECT product, channel, sum of (promo_flag, units_sold), percentage of (total promo_flag units / total units_sold) and then GROUP BY product, channel ORDER by percentage DESC LIMIT 20.

Q17. Which products generate the highest margins?
Ans: SELECT product, category, sum of (net_sales, units sold), average of (margin_pct) then GROUP BY product, category ORDER BY average margin_pct DESC LIMIT 20.

Q18. Which products generate high revenue but relatively weak margins (<38%)?
Ans: Same as Q17, just add HAVING (avg margin_pct) < 0.38 before ORDER BY sum of net_sales clause.

Q19. Which products experience the highest stockout exposure?
Ans: Quite same as Q16, here you need to find sum of 'stock_out_flag' and the percentage of (total stock_out_flag/all record count) then finally ORDER BY percentage DESC.

Q20. Is stockout exposure concentrated among particular stores & cities?
Ans: Same as Q19, just select extra city, store_id field in the query and group accordingly.

Q21. Which products combine high demand with high volatility?
Ans: Demand and Volatility are 2 important measures in supply chain management. Demand is obtained by STDDEV (cumulative units_sold) which gives us a value that represents how much sales/demand can actually vary from average of (cumulative units_sold)
And volatility (or Coefficient of Variation, CV) represents the speed/rate of changes may occur in actual demand relative to average unit sales.
To find demand factor related to products: SELECT sku_id, sku_name, sum of (units_sold), average of (units_sold), STDDEV (units_sold) and then GROUP BY sku_id, sku_name ORDER BY sum of units_sold DESC.
To find CV: SELECT sku_id, sku_name, average of (units_sold), STDDEV (units_sold), CV as [STDDEV (units_sold)/ AVG (units_sold) and then GROUP BY sku_id, sku_name HAVING (avg units_sold > 0) ORDER BY CV DESC.

Q22. How do demand classifications (ABC) and demand-volatility classifications (XYZ) compare?
Ans: To find ABC revenue, do like steps in Q5 (without the limit part). This will generate revenue table for all of the products available (100+ SKU) in DESC order. Now, revenue of top 80% list = A, next 15% = B and the last 5% = C
To find XYZ, do the steps like in Q21 to find a table of CV, of course without any limit. Then according the results, categorize in such way – X = low variability, Y = medium variability, Z = high variability.

Q23. Where are the largest commercial growth and improvement opportunities?
Ans: Opportunity 1 – High revenue products (get answer from Q5)
Opportunity 2 – Highly sold products with stockouts (get answer from combination of Q5 and Q19)
Opportunity 3 – High revenue but low margin (get answer from Q18)